

Mubarak Al Hamadi

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Career Summary

Mechanical Engineering graduate and MSc student with practical exposure to industrial compressors, prototype-scale systems, vapor-compression refrigeration, sensor-based monitoring, and experimental data analysis. Experienced in service operations, research and development, technical documentation, and multidisciplinary coordination, with interests spanning industrial systems, energy, oil and gas, aerodynamics, utilities, and mechatronics.

Education

Master of Science, Abu Dhabi University (Abu Dhabi Campus), Sep 2025 – June 2027
(expected)

Major: Mechanical Engineering with thesis

Related Coursework:

- Project Management
- Advanced Mathematics and Applied Statistics
- Finite Element Applications

Bachelor of Engineering, Abu Dhabi University (Abu Dhabi Campus), Feb 2021- Aug 2025.

- Major: Mechanical Engineering
- GPA: 3.926/4

Related Coursework:

- Control Systems
- Hydraulics & Pneumatics
- Introduction to Mechatronics
- AI in Mechatronics
- Fluid Mechanics
- Machine Design
- Aerodynamics
- Introduction to Business

High School Pass, Al Nukhba School, Sep 2019

- Advanced Stream
- Score: 91.1%

Experience

Research Assistant – Atmospheric Water Harvesting: Sep 2025–Present

- Supported an externally funded atmospheric water harvesting research project funded by Mubadala Muwakaba and Dubai Future Foundation
- Worked on a prototype-scale hybrid AWH system integrating vapor-compression refrigeration, MOF-assisted sorption, thermoelectric cooling, and sensor-based monitoring
- Assisted in experimental testing, performance analysis, and technical documentation under indoor and outdoor operating conditions
- Evaluated system performance through in-depth data logging analysis
- Contributed to technical reports, posters, proposal material, competition submissions, and manuscript-level documentation
- Provided technical guidance to capstone students working on the AWH prototype
- Coordinated research-related administrative tasks, including event organization, submission files, version tracking, supervisor approvals, and stakeholder communication.
- Co-author, “The Innovative and Smart Solar-Powered Atmospheric Water Harvesting System for Clean and Reliable Water Solutions,” in *Artificial Intelligence in Sustainable Energy and Environmental Management*, Springer, 2026

Future17 SDG Challenge Program – Industry Project Mentor | QS / Ekak Innovations – Feb 2026 – Apr 2026

- Mentored a multidisciplinary international team through a 12-week industry challenge with Ekak Innovations
- Managed milestone planning, partner communication, team coordination, mock presentation preparation, and final deliverable review
- Guided documentation and proposal development involving IoT sensors, operational dashboards, AI workflows, and warehouse-management concept

Internship – Atlas Copco Middle East Services: Jun–Aug 2025

- Attended inductions on safety, company vision, and service operations
- Diagnosed and troubleshoot mechanical & electrical faults in compressors
- Conducted oxygen purity testing and membrane replacement in NGUs
- Assisted in installation and reassembly of compressors, pipelines, and control units
- Performed oil checks, filter replacements, and generator fuel pump service
- Participated in site visits to observe industrial applications

AI-Driven Lower Limb Exoskeleton for Mubadala Research – Sep 2024 – Jul 2025

- Researched, designed, and prototyped an AI-powered rehabilitation exoskeleton
- Developed intelligent control systems for improved mobility
- Contributed to a funded research project supported by Mubadala Muwakaba
- Supported technical documentation, prototype development, and project presentation material

Additional Projects & Training

Building Harvesting Rover | Prototype Project: Jun 2023

- Designed, built, tested, and presented a prototype rover for mechanized date and oil palm fruit harvesting.
- Co-author, “Automation and Mechanization of Oil Palm Harvesting Using Drones and Rovers,” Springer, 2026.

Hackathon Competition Organizer: May 2023

- Coordinated participants, provided instructions, supported competition flow, and validated submitted projects.

Research Skills Workshop: May 2023

- Developed research writing, report-structuring, peer-review, and technical presentation skills.

MATLAB for Engineering Applications: Solving Differential Equations: Nov 2024

- Applied MATLAB programming and debugging to solve engineering differential equations.

Heat Treatment Course | Steel University: Mar 2023

- Completed online training in heat-treatment fundamentals. Code: MAN01092.

Skills

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|---------------------------|-----------------------------|----------------------------|
| • Research Writing | • Microsoft (Excel, Word) | • Ansys Software |
| • Flexibility | • CAD (AutoCAD, Inventor) | • Problem Solving |
| • Technical Communication | • Fritzing Software. | • Teamwork & Collaboration |
| • Basic Welding Skills | • MATLAB (Simulink) | • 3D printing skills |
| • Machinery | • ADAMS Software | • Adaptability |
| • Fluid SIM Software | • COMSOL Software | • Video Editing |
| • PCB Design | • Lean Six Sigma (Beginner) | • AI skills |
- Arabic & English (Speaking, Writing, Listening)